

HANDS - ON RAG BOT & AGENTIC AI WORKSHOP

Build an EcoTourism RAG Chatbot + Agent Demo



RAJESH BOPPANA

- MicroChip - Solutions Architect - Edge AI, ML, Gen & Agentic AI
- Ex Solutions Architect @ JPMorgan Chase & @StateFarm
- City Colleges of Chicago (Adjunct Professor)

Education:

- MIT Executive education in AI & Business Strategy(USA)
- Honored student in Master of computer Applications(India)

Linkedin

- **<https://www.linkedin.com/in/rajesh-boppa-551147188/>**



RAG DEMO

EcoTourism AI Assistant (Lightweight)

Ask about sustainable travel, eco-destinations, or green tourism tips!

 Chatbot

What is the top tourist season with the most visitors?

summer

Your question:

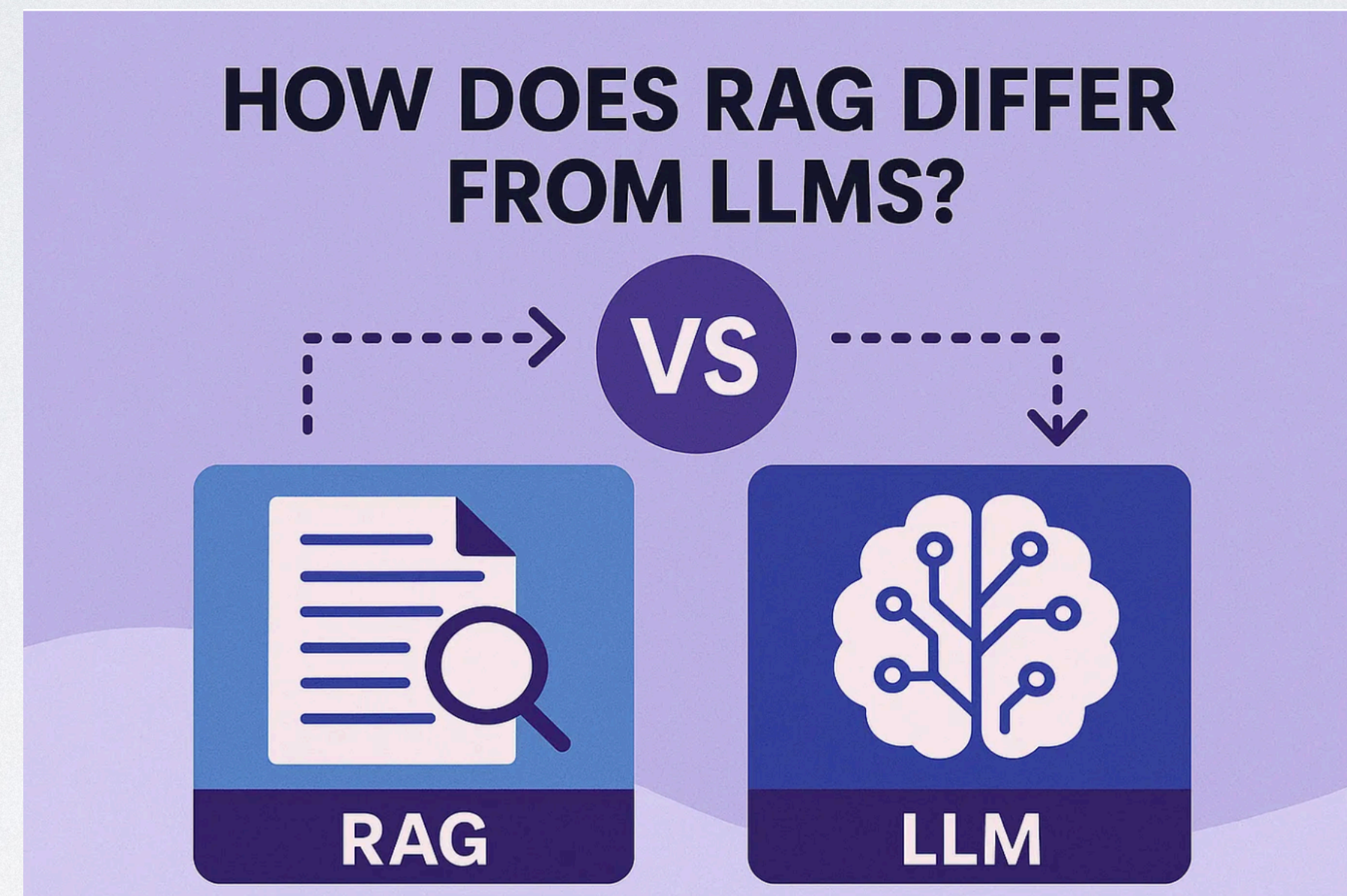
What is the top tourist season with the most visitors?

- <https://huggingface.co/spaces/rajeshboppana/EcoTourismBot>
- What is the top tourist season with the most visitors?

WHAT WE ARE GOING TO DO TODAY ?

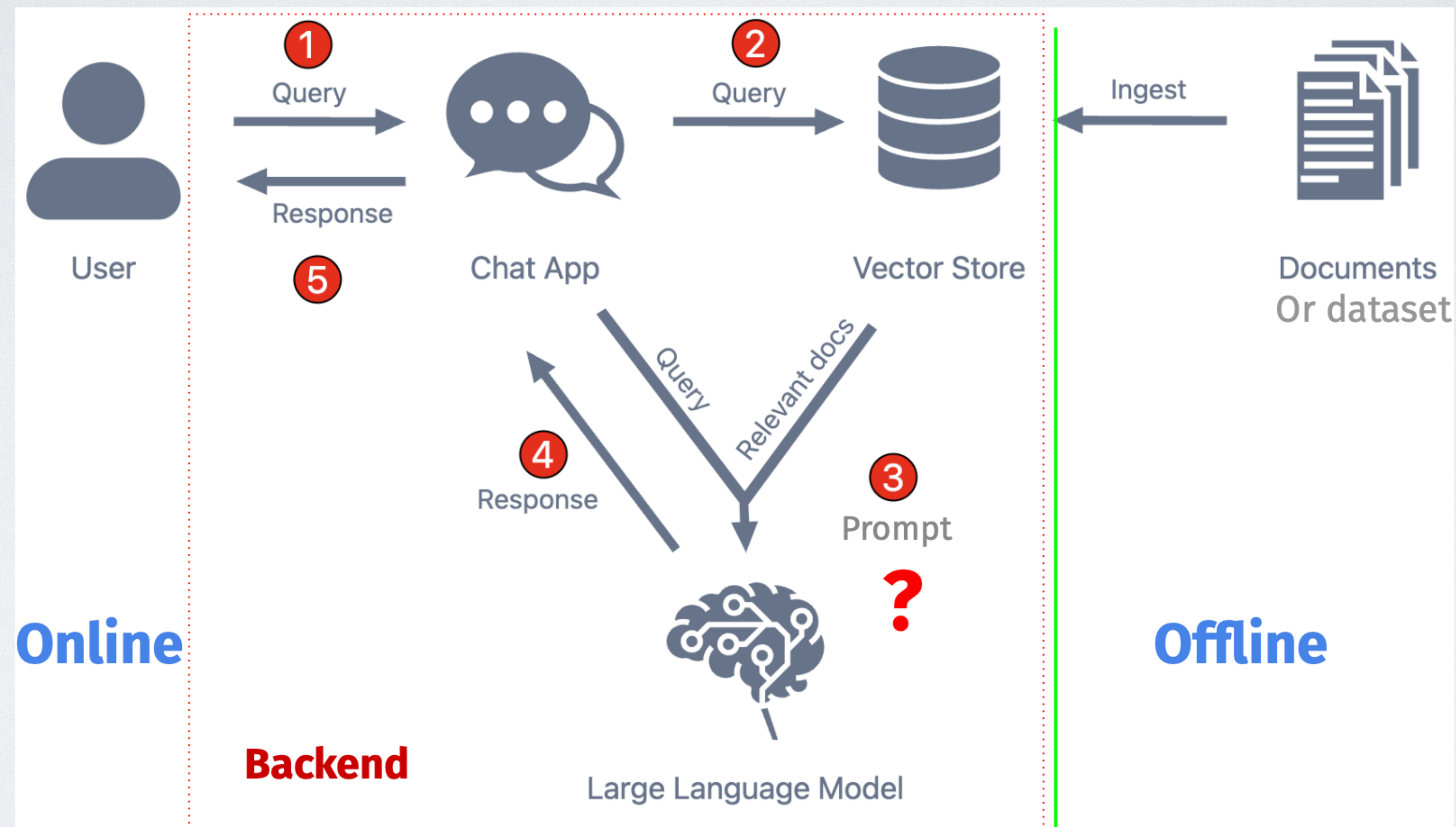
- Prepare Data
- Build Vector Store
- Wire RAG Pipeline
- Build Chatbot
- Create UI
- Test Queries

WHY RAG IS IMPORTANT



“RAG reduces hallucinations by grounding the LLM in real documents.”

RAG ARCHITECTURE



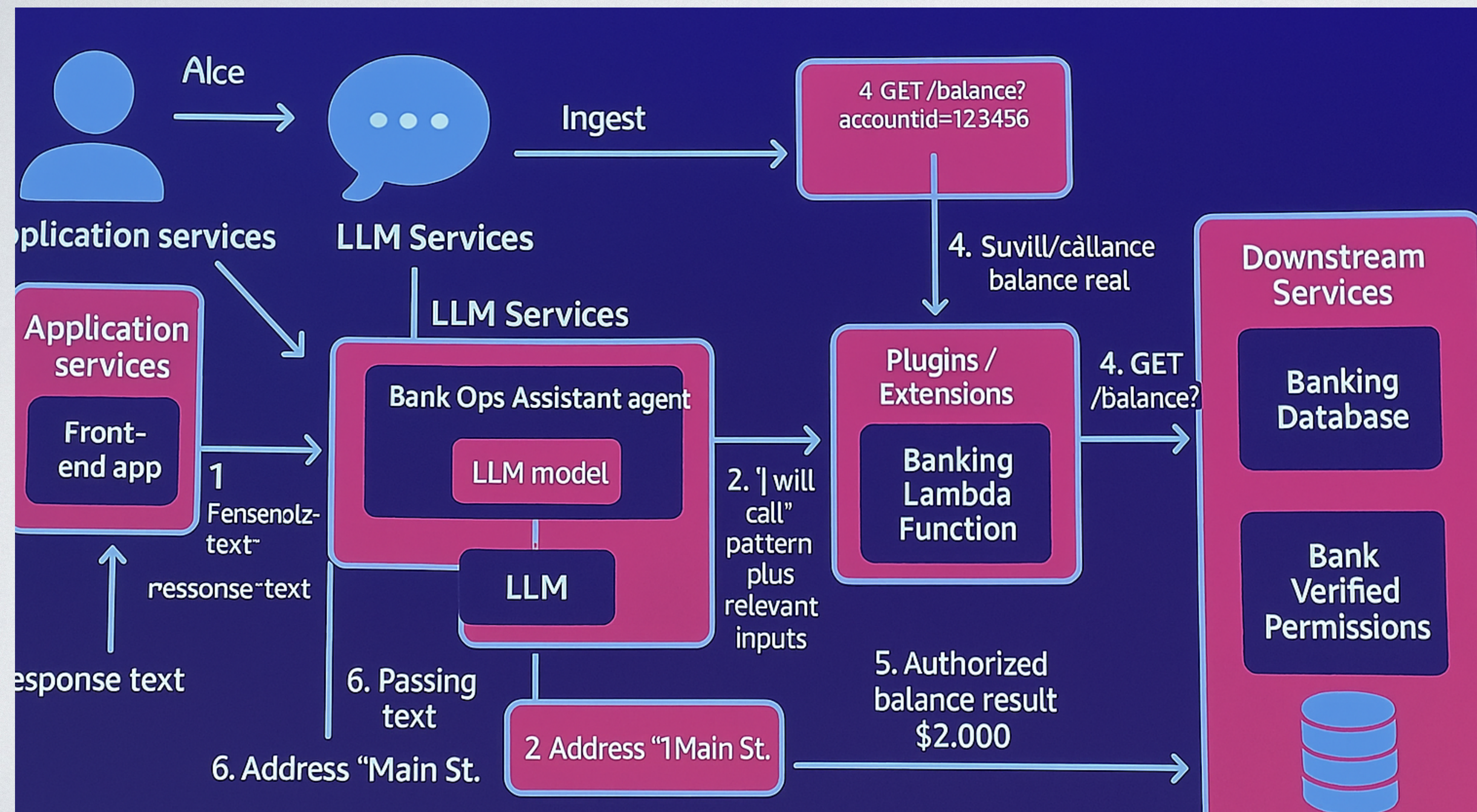
HANDS ON LAB

RAG EVALUATION

- Helpfulness
- Completeness
- Correctness
- Harmfulness (Responsible AI)



HOW TO SECURE A RAG / OWASP



QUESTIONS

AGENTIC AI DEMO

```
#  Sample user input request  
user_input = 'book 3PM slots for Dr. Lee'
```

```
#  Invoke the AI agent to process the request and execute the SQL operation  
output = agent_executor.invoke({"input": user_input})
```

I will now proceed with querying the database to book 3 PM slots for Dr. Lee.

```
[(23, 'Linda Lee', 'Dermatology', 8855162140)]
```

```
Invoking: `sql_db_query` with `{'query': "DELETE FROM slots WHERE doctor_id = 23 AND datetime = '2025-03-01 3:00 P  
M'"}`
```

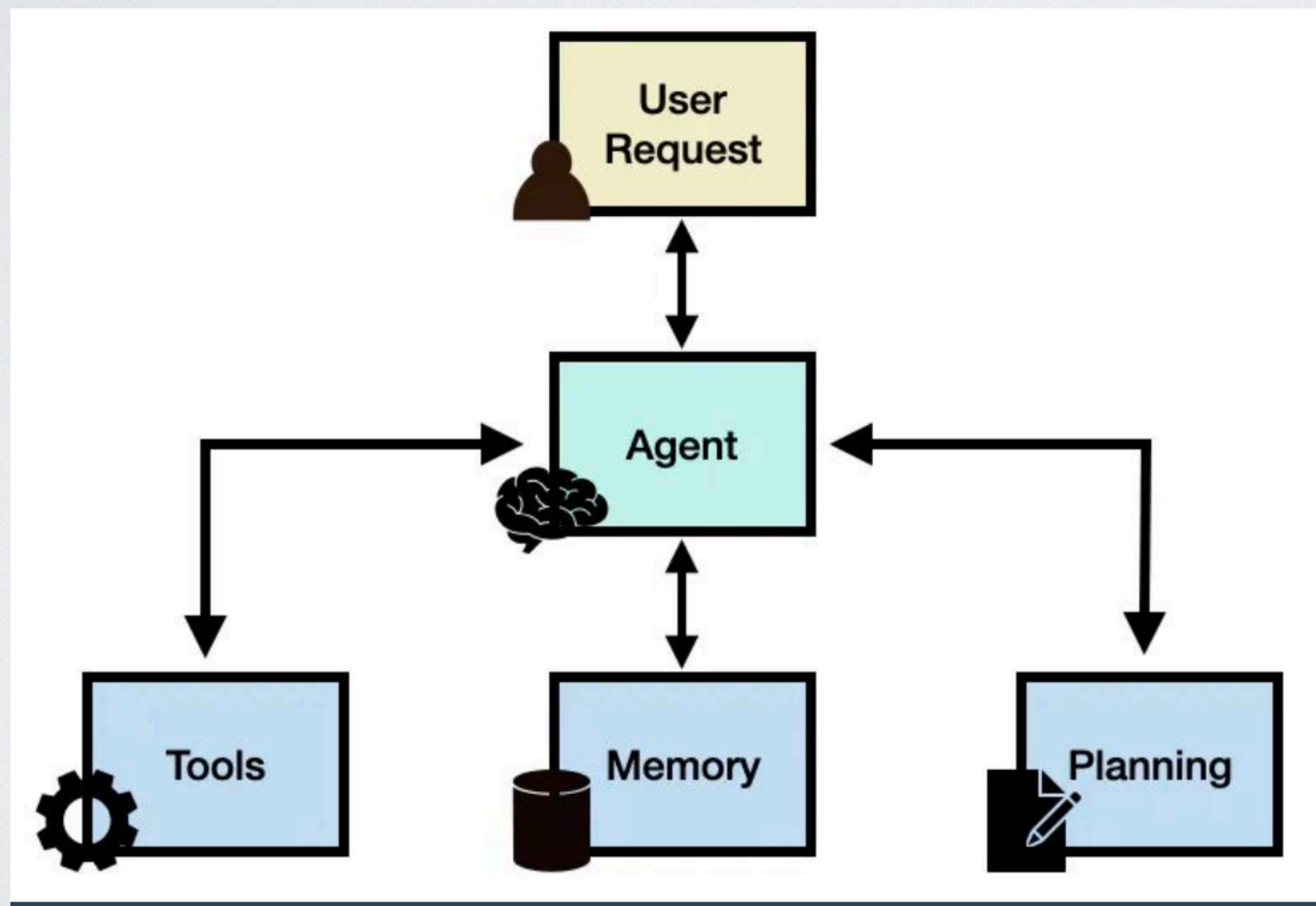
I have successfully booked the 3 PM slot for Dr. Linda Lee. The slot has been reserved.

> Finished chain.

WHY WE NEED AN AGENT

- Take Actions(Book an appointment, Book a flight etc)
- Make Decisions(Choose the best route)
- Interact with the real world.("Book a table at an Italian restaurant near me after 7 PM.")

AGENT ARCHITECTURE



Planner

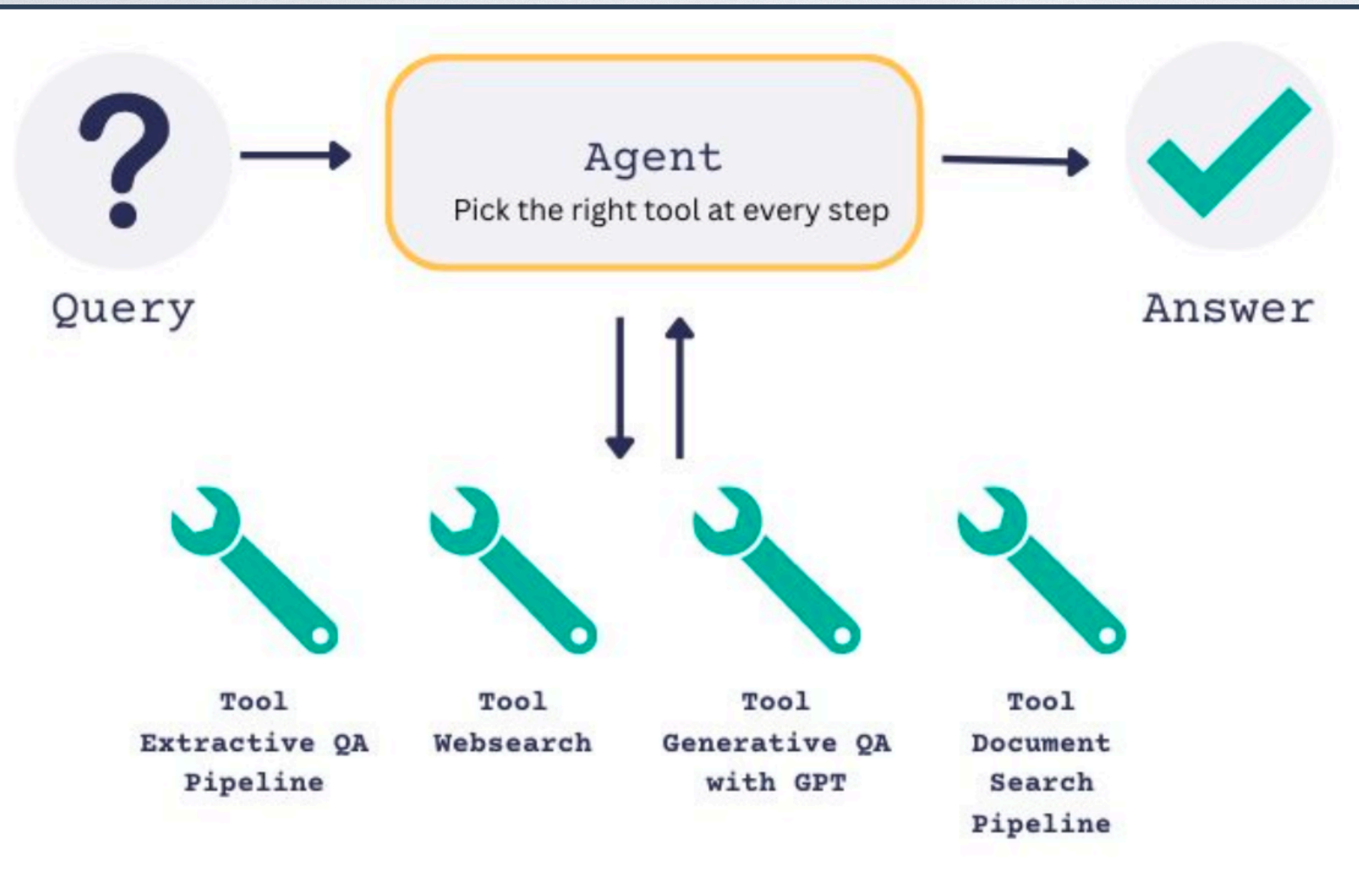
Tools

Memory

LLM

Executor

VARIOUS TYPES OF TOOLS THAT AGENT CAN CONNECT



Organize tools by category:

- Search
- Database
- API
- Utility functions

HANDS ON LAB

QUESTIONS